

Vaaram Magazine Web Platform Architecture, Operations & Admin Guide

An exhaustive walkthrough of how the platform operates, how each public page functions, free-tier storage economics and egress limits, step-by-step administrator publishing instructions, and final GoDaddy domain delegation requirements.

EXECUTIVE SUMMARY FOR CLIENT LEADERSHIP

PLATFORM STATUS: 92% COMPLETE (ENGINE COMPLETE & PRODUCTION OPERATIONAL)

Vaaram Magazine has been engineered as a high-performance weekly digital publication and local advertising platform for Canada. The public reader experience, high-fidelity canvas flipbook reader, automatic and custom thumbnail generators, house advertising engine, rate-limited lead capture, and full administrative portal are **100% complete and verified**.

The remaining **8%** consists purely of final external business configurations: delegating GoDaddy DNS access for **vaaram.ca**, setting up email inbox hosting for **contact@vaaram.ca**, replacing placeholder phone numbers, and publishing the real inaugural Issue #01.

AUTHENTICATION CREDENTIALS

OFFICIAL WEBSITE ADMINISTRATOR CREDENTIALS:

- **Admin Portal URL:** <https://vaaram-magazine.vercel.app/admin/login>
- **Admin Email:** contact@vaaram.ca
- **Admin Password:** Vaaram27#

Note: The login screen also features an email OTP fallback button if password bypass is ever temporarily disabled.

CORE FRAMEWORK	EDGE DEPLOYMENT	DATABASE & AUTH	MEDIA STORAGE
Next.js 14 App Router	Vercel Edge Network	Supabase PostgreSQL + RLS	Supabase S3 Bucket ('media')
READER ENGINE	ADVERTISING SLOTS	CANADIAN COMPLIANCE	CURRENT COMPLETION
HTML5 Canvas PDF.js	Leaderboard, Mid, Feature	CASL & PIPEDA Compliant	92% (READY FOR DELEGATION)

The Reader Experience – Every Page Explained

The public site is designed around speed, elegance, and zero friction: readers never hit paywalls or forced logins.

- 1. Homepage (/):** Features the real-time masthead with ticker announcements, current weekly edition hero showcase, paid leaderboard slot, category directory (Businesses, Services, Property, Jobs, Offers, Community), the 5-step advertiser process, previous editions carousel, total page count statistics, and the email newsletter signup band.
- 2. Archives (/archives):** A comprehensive historical catalogue. Allows readers to browse past editions via 3 distinct views: visual cover cards, chronological list, and publication calendar. Built-in instant client-side filtering allows searching by date or keyword with zero lag.
- 3. Issue Reader (/archives/[slug]):** The core magazine viewing engine. Unlike competitors who rely on clunky browser iframes that break on mobile devices, Vaaram uses a custom **HTML5 Canvas PDF worker**. It renders pages crisply on iOS Safari, Android Chrome, and Desktop with page flipping, zoom controls, fullscreen mode, and a direct high-speed PDF download trigger.
- 4. About Page (/about):** Articulates the magazine's commercial philosophy — a publication intentionally made of curated local advertising. Features the five-step advertising lifecycle and distribution principles.
- 5. Contact & Booking (/contact):** The primary conversion gateway. Displays direct telephone, WhatsApp deep-link, and office address. Houses the interactive quote request form with server-side rate limiting and spam honeypot filters, followed by structured Google-indexed FAQs.
- 6. Privacy Policy & Terms (/privacy, /terms):** Written specifically for Canadian compliance (CASL and PIPEDA), guaranteeing reader privacy, zero tracking profiles, and strict one-click unsubscribe adherence.

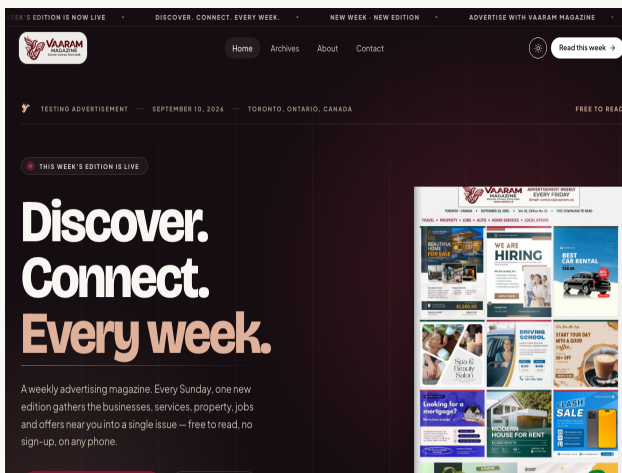


FIG: Homepage: Masthead, Weekly Hero & Reader Entry

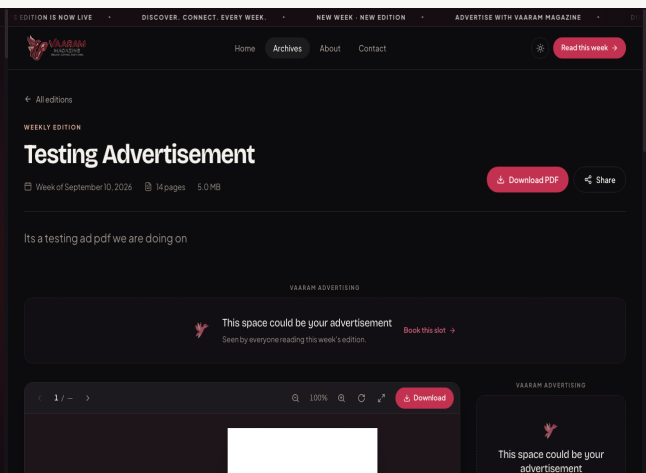


FIG: Interactive Reader: Mobile Canvas Flipbook Engine

Understanding Free Tier Limits & Future Scaling

Transparent analysis of database and storage quotas, egress calculations, and 4 zero-cost scaling strategies.

What is the Supabase Free Tier?

The website is currently deployed with **Supabase Storage** as its media backplane. The Supabase free tier provides **500 MB of PostgreSQL database storage** (enough for over 50,000 editions, ad records, and enquiries), **1 GB of free file storage**, and an egress bandwidth cap of roughly **2 GB to 5 GB per month**.

EGRESS BANDWIDTH MATHEMATICS

UNDERSTANDING 'EGRESS BANDWIDTH' IN PLAIN ENGLISH:

- **What is Egress?** Egress is the amount of data transferred out from the storage servers whenever a reader downloads or reads an edition.
- **The Formula:** $\text{Monthly Egress} = (\text{Average PDF Size}) \times (\text{Number of Full Reads} / \text{Downloads})$
- **Current Reality:** A standard 12-page magazine exported at standard print quality is ~4 MB.
- On a 2 GB free egress plan: $2,000 \text{ MB} \div 4 \text{ MB} = \sim 500 \text{ full edition reads per month}$.
- If your publication reaches 1,500 monthly readers, free egress bandwidth will be surpassed.

4 Proven Strategies to Bypass Egress Limits (100% Free / No Credit Card)

STRATEGY	HOW IT WORKS	BANDWIDTH IMPACT	EFFORT
1. 150 DPI Optimization	Export magazine PDFs from Canva/InDesign at 150 DPI (Web Quality) instead of 300 DPI (Heavy Print). Reduces size from 6 MB to ~1.8 MB.	2.5× Capacity Increases free reads from 500 to 1,250/mo.	Zero code (Export setting)
2. Cloudflare Free CDN Proxy	Point the domain through Cloudflare's free tier. Static PDF files are cached at global edge data centres; repeat reads are served from Cloudflare cache rather than hitting Supabase egress.	90% Egress Shield Over 90% of repeat reads cost zero Supabase bandwidth.	15 minutes (DNS toggle)
3. Two-Tier GitHub Releases	Keep 'This Week's Current Edition' on fast Supabase Storage. When an edition is archived, offload the PDF to a GitHub Release asset (unlimited free bandwidth via jsDelivr CDN).	Unlimited Free Infinite archive downloads without paying a penny.	Moderate (Admin button)
4. Internet Archive (archive.org)	Internet Archive is a free global library dedicated to historical periodicals. Old issues can be hosted permanently with unlimited public downloads and high-speed delivery.	Permanent Free Historical preservation forever.	Zero code (Copy URL)

Sign In, Authentication & Dashboard Overview

How to access the publishing desk, monitor platform vital signs, and navigate administrative modules.

Accessing the Portal: Navigate to <https://vaaram-magazine.vercel.app/admin/login>. The login screen is clean and uncluttered. Enter your administrator email and password to enter directly.

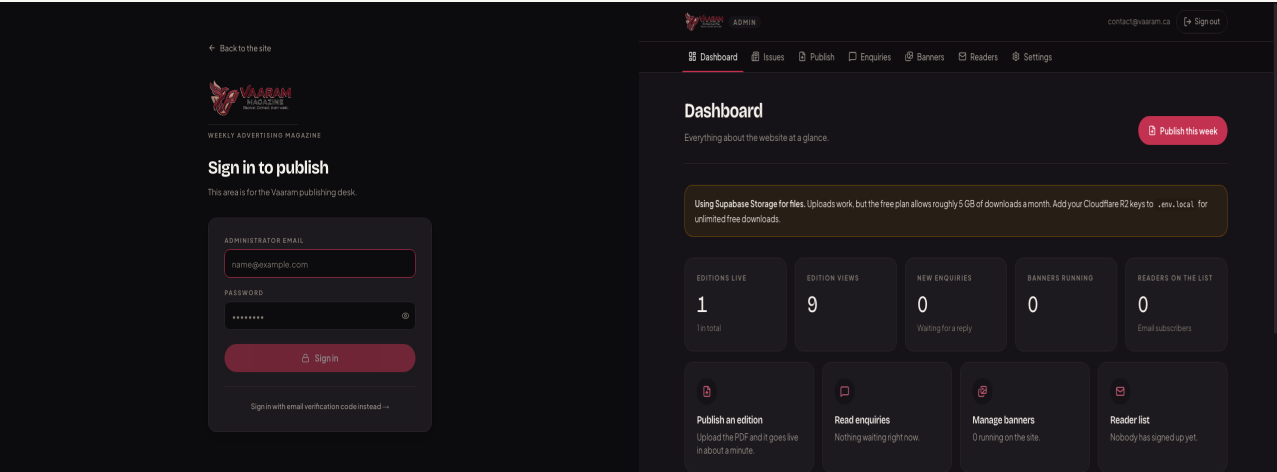


FIG: Admin Login: Direct Password Authentication with OTP Fallback

FIG: Admin Dashboard: Real-Time Publications, Views & Inquiries

Navigating the Dashboard Metrics

Upon successful authentication, the admin arrives at the central command dashboard containing 5 real-time telemetry tiles:

- **Editions Live:** Total number of published editions currently accessible to readers on the homepage and archives.
- **Edition Views:** Real-time counter of total flipbook sessions and reads across all publications.
- **New Enquiries:** Number of unreviewed advertiser inquiries received through the contact form.
- **Banners Running:** Count of active paid sponsor banners currently displaying on the public website.
- **Readers on the List:** Total confirmed email subscribers signed up to receive weekly edition drops.

Publishing a New Weekly Edition (/admin/issues/new)

Step-by-step instructions for uploading print-ready PDFs, thumbnail generation, and instant release.

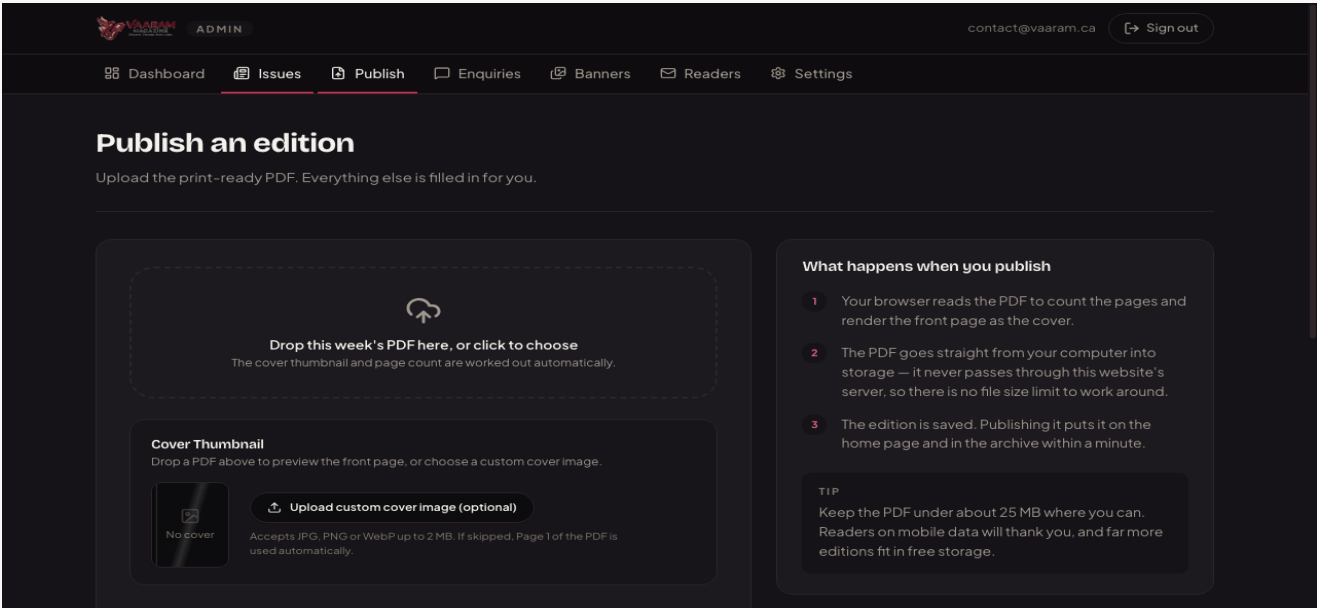


FIGURE: The Publishing Interface: PDF Dropzone, Custom Thumbnail Uploader, and Auto-Extraction Engine

Complete Example Posting Walkthrough

Follow these exact steps every week (normally every Sunday morning) to publish the new edition:

- **Step 1 — Drop the PDF:** Drag your weekly magazine PDF into the main dropzone (e.g. *vaaram-issue-01.pdf*). The browser directly verifies file integrity.
- **Step 2 — Cover Thumbnail Selection:**
 - *Option A (Automated):* Leave it empty. The system's built-in canvas engine will automatically render Page 1 of the PDF as the cover.
 - *Option B (Custom Thumbnail - Recommended):* Click '**Upload custom cover image**' and select a dedicated high-resolution JPEG/PNG (1200×1600 px). This ensures razor-sharp artwork across retina displays.
- **Step 3 — Input Title & Edition Details:**
 - *Edition Title:* Vaaram Magazine — Issue 01 (Inaugural Edition)
 - *Issue Number:* 1
 - *Publication Date:* 2026-09-14
 - *Description:* Inaugural Canadian edition featuring local community businesses, legal & financial services, automotive sales, and weekly classifieds.
- **Step 4 — Status Toggle:** Keep '**Publish immediately**' checked to launch live on the homepage hero, or uncheck it to save as a private Draft for review.
- **Step 5 — Publish:** Click '**Publish edition**'. Within seconds, the PDF streams into storage, the database updates, and the homepage hero displays the new edition!

ADMIN WORKFLOW 2

Editing Existing Issues & Digital Banner Management

How to correct published editions, upload custom thumbnails post-launch, and sell web banner ads.

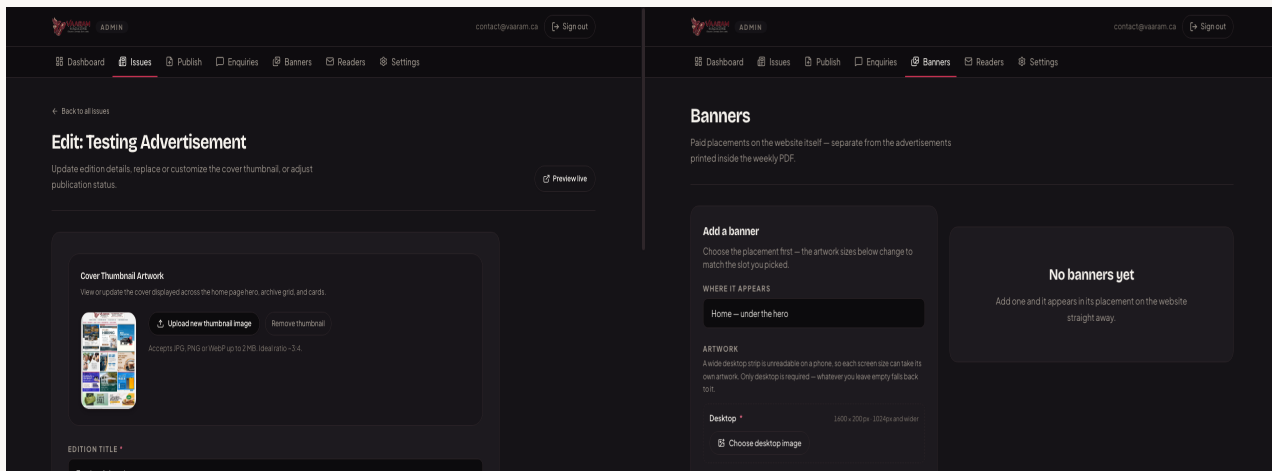


FIG: Issue Editing: Change Titles, Replace PDF, or Upload Custom Thumbnail

FIG: Banner Manager: Sell Leaderboards & Mid-Page Digital Ad Placements

How to Edit an Existing Issue (/admin/issues/[id]/edit)

If you ever spot a typo in a published issue, need to update the date, or want to replace the PDF file with an updated ad proof:

1. Go to **Admin** □ **Issues** and click **Edit** next to the publication.
2. You can freely edit the **Title**, **Issue Number**, **Edition Date**, or **Description**.
3. To replace the cover artwork, click '**Upload new thumbnail image**'.
4. To replace the entire PDF file, drop the new PDF into the replacement zone.
5. Click '**Save changes**'. The live website updates immediately without changing the public link or losing view statistics.

How to Manage Digital Advertising Banners (/admin/banners)

Vaaram sells advertising space on the website itself in addition to print ads inside the PDF:

• Where Banners Appear:

1. Home — under the hero (High-impact 1600×200 desktop leaderboard).
2. Home — middle of the page (Mid-page reader engagement placement).
3. Home — feature panel (Full-width showcase sponsorship).
4. Archives — sidebar (Persistent alongside the back-issue index).

• **Input Details:** Select placement, upload Desktop artwork (Tablet/Mobile optional), set the Sponsor Name, and add the Target Link URL (e.g. <https://sponsor-website.ca>).

• **The Branded House Ad Engine:** If no paid sponsor is currently booked for a placement, the site automatically renders a tasteful Vaaram House Ad (*"This space could be your advertisement — Book this slot □"*). **The site never shows empty or broken boxes.**

ADMIN WORKFLOW 3

Advertiser Inquiries & Newsletter Subscribers

Managing inbound client leads, quotes, and weekly reader newsletter distribution.

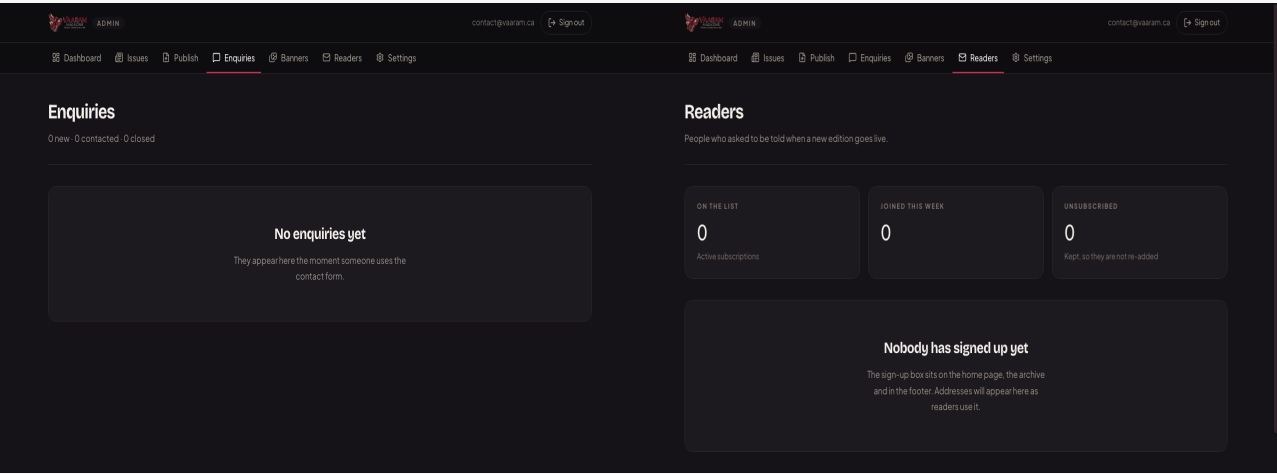


FIG: Inquiries Inbox: Review Inbound Quote Requests & Advertiser Leads

FIG: Reader Subscriber List: Verified Email List for Weekly Drops

Managing Advertiser Enquiries (/admin/enquiries)

Whenever a prospective business submits the enquiry form on the Contact page, the lead is safely recorded in the database:

- **Captured Information:** Full Name, Contact Telephone, Optional Email, Selected Category, Subject Line, and Detailed Message.
- **Built-in Security:** Inbound forms are shielded by a sliding-window rate limiter (max 6 requests per IP per 10 minutes) and a hidden honeypot trap field that silently swallows automated malicious bots without impacting genuine Canadian business owners.

Managing Newsletter Subscribers (/admin/subscribers)

Readers can enter their email on the homepage to be notified the moment Sunday's edition drops:

- **Security & Privacy:** Subscribers are secured using Supabase Row-Level Security (RLS). Public users can only write their own email; they can never read or scrape other people's addresses.
- **Unsubscribe Honesty:** When a user asks to be removed, their status is set to inactive rather than deleted. This ensures future contact imports never accidentally re-subscribe an address that opted out.

The Remaining 8% – Client Requirements & GoDaddy Delegation

Specific credentials and account delegations needed from the client to execute the official Canadian public release.

To graduate the website from **92% Complete** to **100% Officially Launched**, the client must complete the following 4 actionable steps:

ACTION 1: GODADDY DELEGATION

ACTION ITEM 1: GODADDY DOMAIN DELEGATION (CRITICAL)

To connect **vaaram.ca** and **www.vaaram.ca** to Vercel's global edge network, the developer team requires DNS management access. You do not need to share your personal GoDaddy password. You can simply add our developer account as a **Delegate Access User**:

- **Delegate Email Address:** **nivassri183@gmail.com**
- **Permission Level:** Products & Domains (or 'Domains Only')

How to Delegate in GoDaddy (Takes 2 Minutes):

1. Log in to your GoDaddy Account.
2. Click your name in the top-right corner ☐ select '**Account Settings**'.
3. Click '**Delegate Access**'.
4. In the '*People who can access my account*' panel, click '**Invite to Access**'.
5. Enter Name: **Nivas** and Email: **nivassri183@gmail.com**.
6. Select '**Products & Domains**' and click '**Invite**'.

Once invited, we can instantly configure the DNS A-records and SSL security certificates.

ACTION 2: EMAIL INBOX HOSTING

ACTION ITEM 2: EMAIL INBOX SETUP (contact@vaaram.ca)

Currently, **contact@vaaram.ca** is the declared brand email across the website and admin portal. However, **the team does not have access to an active email inbox for contact@vaaram.ca**.

- **Requirement:** Please confirm where **contact@vaaram.ca** is hosted (e.g. Google Workspace, Microsoft 365, or GoDaddy Email Essentials).
- If an inbox has not yet been purchased, we can either configure a free email forwarder in GoDaddy to forward incoming emails to your personal inbox, or help you connect Google Workspace.

ACTIONS 3 & 4: PHONE & CONTENT

ACTION ITEMS 3 & 4: REAL PHONE NUMBER & INAUGURAL ISSUE #01

- **Real Phone & WhatsApp:** Provide your official Canadian telephone number and WhatsApp business number to replace the temporary 555 placeholder.
- **Official Issue #01 PDF:** Provide the finalized Week 1 magazine PDF and cover image to upload via the admin portal before opening to the public.